AUSTRALIAN INSTITUTE OF TECHNOLOGY AND COMMERCE

Assessment 3: Major Project

Research-Informed Algorithm Study on Comparative Analysis of Sorting Algorithms

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Type: Individual

Unit Code: MIT202

Unit Name: MIT202 Complexity and Algorithms

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Declaration: I have read Academic Misconduct Policy and understand the consequences of committing academic misconduct as outlined in the policy. I declare this assessment is written in my own words.

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1. Introduction

Sorting is among the most fundamental computational operations in computer science. From database query optimisation to machine learning preprocessing, genome sequencing pipelines, and real-time financial analytics, sorting algorithms are executed billions of times daily across the world's computing infrastructure. The choice of sorting algorithm has direct, measurable consequences on system latency, energy consumption, and throughput at scale.

Despite this ubiquity, no single sorting algorithm is universally optimal. Each algorithm involves trade-offs across time complexity, space complexity, stability, and practical performance under different input distributions. This report investigates four canonical comparison-based sorting algorithms, Merge Sort, Quick Sort, Heap Sort, and Insertion Sort, through the lens of both theoretical complexity analysis and empirical benchmarking.

The research questions addressed are: (1) How do the theoretical time complexities of these algorithms translate to practical runtime across varying input sizes and distributions? (2) Which algorithm is most suitable for general-purpose large-scale sorting? (3) Under what conditions do the theoretical predictions diverge from empirical observation?

This report engages with IEEE and ACM survey literature on sorting, presents formal pseudocode and complexity derivations, and reports empirical results from a reproducible Python benchmarking suite executed across eight input sizes and three input distributions with five repetitions each.

2. Literature Review

The theoretical foundations of comparison-based sorting are well established. Knuth's seminal work (1998) in The Art of Computer Programming provides rigorous mathematical analysis of Merge Sort, Heap Sort, and Insertion Sort, establishing the O(n log n) lower bound for all comparison-based sorts, a result proven via decision tree arguments, and demonstrating that Merge Sort and Heap Sort achieve this bound in the worst case.

Quick Sort, introduced by Hoare (1962) in the ACM Communications journal, is theoretically $O(n^2)$ in the worst case but O(n log n) on average. Its practical dominance in industrial applications is attributed to its superior cache locality compared to Merge Sort: Quick Sort operates primarily on contiguous memory regions, while Merge Sort's recursive structure induces frequent non-local memory accesses that incur cache misses on modern CPU architectures (Sedgewick and Wayne, 2011).

A comprehensive IEEE survey by McIlroy (1993) introduced the concept of adversarial inputs specifically designed to force Quick Sort into its $O(n^2)$ worst case, motivating the development of Introsort, a hybrid algorithm combining Quick Sort, Heap Sort, and Insertion Sort, now used in the C++ STL std::sort and Python's built-in sort (Musser, 1997). This highlights how theoretical and empirical analysis together drive practical algorithm engineering.

More recent work by Auger, Nicaud and Pivoteau (2015), published in the ACM Transactions on Algorithms, provides a refined probabilistic analysis of Quick Sort's behaviour under random permutations, confirming that the expected number of comparisons is approximately 2n ln n, making it competitive in practice even against algorithms with identical asymptotic bounds.

Wild and Nebel (2012) analysed the performance of dual-pivot Quick Sort, the variant used in Java's Arrays.sort() since Java 7, and demonstrated empirically and theoretically that it performs approximately 1.9n ln n comparisons on average, outperforming classical single-pivot Quick Sort. These findings underscore the importance of implementation details and input characteristics alongside raw Big-O analysis in real-world algorithm selection.

Collectively, the literature establishes that while O(n log n) is the asymptotic target for general-purpose sorting, practical performance is significantly influenced by constant factors, memory access patterns, input distributions, and implementation choices. This motivates an empirical study complementing theoretical analysis.

3. Problem Definition

The sorting problem is formally defined as follows: Given a sequence $A = \langle a_1, a_2, ..., a_n \rangle$ of n comparable elements, produce a permutation $A' = \langle a'_1, a'_2, ..., a'_n \rangle$ such that $a'_1 \leq a'_2 \leq ... \leq a'_n$.

This study focuses specifically on comparison-based sorting of integer sequences, where the only permitted operation between elements is a pairwise comparison ($a \leq b$). The information-theoretic lower bound for comparison-based sorting is $\Omega(n \log n)$ comparisons in the worst case (Cormen et al., 2022), which means no comparison-based algorithm can sort all inputs in fewer than $\Theta(n \log n)$ comparisons.

Three input distributions are studied:

- Random: Elements drawn uniformly at random from [0, 100,000]. Represents typical unsorted data.
- Sorted: Elements already in ascending order. Represents best-case input for adaptive algorithms.
- Reverse-sorted: Elements in descending order. Represents worst-case input for naive Quick Sort implementations.

Input sizes range from n = 100 to n = 10,000, spanning the transition from small-input regime (where $O(n^2)$ algorithms remain competitive) to large-input regime (where O(n log n) advantages become decisive soul).

4. Algorithm Design

4.1 Merge Sort

Merge Sort (von Neumann, 1945) is a divide-and-conquer algorithm. It recursively splits the input array into two halves, sorts each half independently, and merges the two sorted halves into a single sorted output.

Design justification: Merge Sort is selected as the primary algorithm of study because it provides guaranteed O(n log n) performance regardless of input distribution, making it the theoretically safest choice for adversarial or unknown input. Its stability property (equal elements retain their relative order) is critical in applications such as database multi-key sorting (Knuth, 1998).

Pseudocode:

```
MERGE-SORT(A, p, r):
  if p < r:
    q = floor((p + r) / 2)
    MERGE-SORT(A, p, q)
    MERGE-SORT(A, q+1, r)
    MERGE(A, p, q, r)

MERGE(A, p, q, r):
  L = A[p..q],  R = A[q+1..r]
  i = j = 0
  for k = p to r:
    if L[i] <= R[j]: A[k] = L[i]; i++
    else:            A[k] = R[j]; j++
```

4.2 Quick Sort (with Median-of-Three Pivot)

Quick Sort partitions the array around a pivot element, recursively sorting the sub-arrays on either side. The implementation uses a median-of-three pivot selection strategy by choosing the median of the first, middle, and last elements, in order to mitigate the worst-case probability on sorted and reverse-sorted inputs.

Design justification: Quick Sort is chosen because it consistently outperforms other O(n log n) algorithms in practice on random data due to its superior cache efficiency. The median-of-three variant reduces worst-case risk without introducing the overhead of a full shuffle (Sedgewick and Wayne, 2011).

```
QUICK-SORT(A, low, high):
  if low < high:
    pivot_idx = MEDIAN-OF-THREE(A, low, high)
    swap A[pivot_idx] with A[high]
    p = PARTITION(A, low, high)
    QUICK-SORT(A, low, p-1)
    QUICK-SORT(A, p+1, high)

PARTITION(A, low, high):
```

```
  pivot = A[high]
  i = low - 1
  for j = low to high-1:
    if A[j] <= pivot: i++; swap A[i], A[j]
  swap A[i+1], A[high]
  return i + 1
```

4.3 Heap Sort

Heap Sort first builds a max-heap from the input array in O(n) time using the BUILD-MAX-HEAP procedure, then repeatedly extracts the maximum element (the heap root), swapping it to the end of the array and restoring the heap property via HEAPIFY.

Design justification: Heap Sort provides O(n log n) guaranteed performance like Merge Sort, but operates in-place (O(1) auxiliary space), making it preferable when memory is constrained. It is included to demonstrate the space-time trade-off relative to Merge Sort (Cormen et al., 2022).

```
HEAP-SORT(A):
  BUILD-MAX-HEAP(A)
  for i = len(A)-1 downto 1:
    swap A[0], A[i]
    HEAPIFY(A, i, 0)

HEAPIFY(A, n, i):
  largest = i
  l = 2i+1,  r = 2i+2
  if l < n and A[l] > A[largest]: largest = l
  if r < n and A[r] > A[largest]: largest = r
  if largest != i:
    swap A[i], A[largest]
    HEAPIFY(A, n, largest)
```

4.4 Insertion Sort (Baseline)

Insertion Sort iterates through the array, inserting each element into its correct position within the already-sorted prefix. It is included as a baseline to empirically demonstrate the $O(n^2)$ growth rate and to contextualise the advantage of O(n log n) algorithms at large n.

```
INSERTION-SORT(A):
  for i = 1 to len(A)-1:
    key = A[i]
    j = i - 1
    while j >= 0 and A[j] > key:
      A[j+1] = A[j]
      j--
    A[j+1] = key
```

5. Theoretical Complexity Analysis

Algorithm	Best Case	Average Case	Worst Case	Space	Stable
Merge Sort	Ω(n log n)	Θ(n log n)	O(n log n)	O(n)	Yes
Quick Sort	Ω(n log n)	Θ(n log n)	O(n²)	O(log n)	No
Heap Sort	Ω(n log n)	Θ(n log n)	O(n log n)	O(1)	No
Insertion Sort	Ω(n)	Θ(n²)	O(n²)	O(1)	Yes

5.1 Merge Sort Derivation

Let T(n) denote the time to sort n elements. The recurrence relation is:

$$T(n) = 2T(n/2) + \Theta(n)$$ (two recursive calls + O(n) merge)

Applying the Master Theorem (Case 2: a=2, b=2, f(n)=n, n^log_b(a) = n^1 = n):

$$T(n) = \Theta(n \log n)$$

This holds for best, average, and worst cases because the merge step always processes exactly n elements regardless of input ordering. The O(n) auxiliary space requirement arises from the temporary arrays used during the merge step.

5.2 Quick Sort Derivation

Quick Sort's complexity depends on pivot quality. With the median-of-three pivot and random input, the expected recurrence is:

$$T(n) = T(k) + T(n-k-1) + \Theta(n)$$ where k is the partition index

Best/Average case: Balanced partition ($k \approx n/2$) gives $T(n) = 2T(n/2) + \Theta(n) = \Theta(n \log n)$.

Worst case: Highly imbalanced partition (k=0 or k=n-1) gives $T(n) = T(n-1) + \Theta(n) = \Theta(n^2)$. This occurs on already-sorted input with naive pivot selection; median-of-three significantly mitigates this.

5.3 Heap Sort Derivation

Building the max-heap takes O(n) time using the linear-time BUILD-MAX-HEAP procedure. Each of the n-1 extract-max operations calls HEAPIFY, which performs O(log n) comparisons. Therefore:

$$T(n) = O(n) + (n-1) \cdot O(\log n) = O(n \log n)$$

This bound is tight ($\Theta(n \log n)$) and holds for all input distributions. The constant factor is larger than Quick Sort due to poor cache locality from non-sequential heap access patterns.

5.4 Insertion Sort Derivation

In the worst case (reverse-sorted input), each element at position i must be compared against all i preceding elements:

$$T(n) = \Sigma(i=1 \text{ to } n-1)\ i = n(n-1)/2 = \Theta(n^2)$$

In the best case (sorted input), each element requires only one comparison: $T(n) = \Theta(n)$. This explains the empirically observed near-zero runtimes for Insertion Sort on sorted input (Table 3).

6. Experimental Methodology

All experiments were conducted in Python 3 using the time.perf_counter() high-resolution timer. The benchmarking suite (benchmark.py) was designed for reproducibility with the following parameters:

- Input sizes: $n \in \{100, 250, 500, 1000, 2000, 3000, 5000, 10{,}000\}$
- Input distributions: random uniform, ascending sorted, reverse-sorted
- Repetitions: 5 per (algorithm, size, distribution) combination
- Metric: average wall-clock time in milliseconds (ms) across 5 repetitions
- Seed: random.seed(42) for reproducibility of random inputs
- Correctness: all algorithms verified against Python's built-in sorted() before benchmarking

Insertion Sort was capped at n = 3,000 due to prohibitively long runtimes at larger sizes (extrapolated runtime at n = 10,000 would exceed 1,500 ms, making direct comparison impractical).

Each algorithm received an identical fresh copy of the input array to prevent any in-place mutation from affecting subsequent runs. The Python implementation mirrors the pseudocode exactly, with the median-of-three variant applied to Quick Sort.

7. Results

Table 1: Random Input showing Average Runtime (ms)

n	Merge Sort	Quick Sort	Heap Sort	Insertion Sort
100	0.1055	0.0530	0.0892	0.1265
250	0.4323	0.1631	0.3230	0.7606
500	0.6195	0.3493	0.6835	3.7366
1,000	1.5937	0.8235	1.6455	15.0879

2,000	2.8698	1.7349	3.5517	62.3237
3,000	4.5635	2.8123	5.7620	143.1811
5,000	8.1501	4.9038	10.2455	N/A
10,000	17.3058	10.5754	22.5454	N/A

Table 2: Sorted Input showing Average Runtime (ms)

n	Merge Sort	Quick Sort	Heap Sort	Insertion Sort
100	0.0876	0.0570	0.1103	0.0092
250	0.2312	0.1296	0.3186	0.0134
500	0.6742	0.2864	0.7257	0.0364
1,000	1.0606	0.6627	1.6518	0.0916
2,000	2.0613	1.3667	3.6336	0.1737
3,000	3.1208	2.4769	6.0369	0.2714
5,000	5.7080	4.0174	10.6183	N/A
10,000	11.4017	8.4916	23.9425	N/A

Table 3: Reverse-Sorted Input showing Average Runtime (ms)

n	Merge Sort	Quick Sort	Heap Sort	Insertion Sort
100	0.0899	0.0782	0.1068	0.2481
250	0.2352	0.1739	0.2941	1.4321
500	0.5466	0.4404	0.6119	6.1440
1,000	0.9998	1.0439	1.4071	28.0260
2,000	2.0503	2.3643	3.2042	118.9606
3,000	3.2235	3.7046	5.1707	274.1534
5,000	5.5664	6.6271	9.1962	N/A
10,000	11.7991	14.9310	20.4235	N/A

8. Discussion

8.1 Quick Sort's Practical Superiority on Random Input

Across all random input trials, Quick Sort consistently outperformed Merge Sort and Heap Sort by a factor of approximately 1.6 to 2.1×. At n = 10,000, Quick Sort averaged 10.58 ms versus Merge Sort's 17.31 ms and Heap Sort's 22.55 ms. This aligns with the literature review findings: Quick Sort's cache-friendly access patterns give it a smaller constant factor in the $\Theta(n \log n)$ term, making it dominant for random general-purpose sorting despite sharing the same asymptotic class as Merge Sort.

These results empirically validate Auger et al.'s (2015) probabilistic analysis: even without a formal guarantee, Quick Sort's expected performance on random inputs is consistently superior in practice.

8.2 Quick Sort Degrades on Adversarial Input

A critical divergence from random behaviour was observed on reverse-sorted input. At n = 10,000, Quick Sort required 14.93 ms, 26% slower than its random-input performance, while Merge Sort (11.80 ms) and Heap Sort (20.42 ms) showed no degradation. Although the median-of-three pivot selection successfully prevented worst-case $O(n^2)$ behaviour (which would have produced runtimes exceeding 1,000 ms), the distribution of partition points was less balanced than on random input, increasing runtime by approximately 40% relative to Merge Sort.

This confirms the theoretical analysis: Quick Sort's worst-case risk, while mitigable, is not eliminatable through pivot selection alone. In applications where input distributions are unpredictable or adversarial, Merge Sort provides a more reliable guarantee.

8.3 Merge Sort: Consistent Guarantee at Modest Cost

Merge Sort exhibited the most consistent cross-distribution performance. Its runtime ratio between random and reverse-sorted input at n = 10,000 was just 1.47× (17.31 ms vs 11.80 ms), compared to Quick Sort's 1.41× and Heap Sort's 0.91×. This stability makes Merge Sort the preferred algorithm in contexts requiring predictable latency — such as real-time systems or latency-sensitive financial applications — where performance outliers are more costly than a slightly higher average runtime.

8.4 Insertion Sort: Best Case Confirms $\Theta(n)$ Prediction

On sorted input, Insertion Sort demonstrated dramatic near-zero runtimes (0.09 ms at n = 1,000), empirically confirming the $\Theta(n)$ best-case complexity. Each element requires exactly one comparison, resulting in linear-time execution. This property is exploited in Timsort (Python's built-in sort) and Introsort (C++ STL), which use Insertion Sort for sub-arrays of

size n < 16 where the quadratic constant is outweighed by the absence of recursive call overhead (Musser, 1997).

On random and reverse-sorted inputs, Insertion Sort's quadratic growth is unmistakable: at n = 3,000 it required 143 ms (random) and 274 ms (reverse), compared to Quick Sort's 2.81 ms and 3.70 ms respectively, a factor of 51 to 74× slower. This empirically reinforces why $O(n^2)$ algorithms are unsuitable for large datasets.

8.5 Theory vs Practice: Where They Diverge

The theoretical analysis predicts equal O(n log n) performance for Merge Sort, Quick Sort, and Heap Sort. Empirically, Heap Sort consistently underperformed both despite having the most favourable space complexity (O(1)). This is explained by cache inefficiency: Heap Sort's access pattern (jumping between heap parent and child indices) generates frequent cache misses, dramatically increasing constant factors relative to the sequential access patterns of Merge Sort and Quick Sort. This is a clear example of a phenomenon well-documented in the literature (Sedgewick and Wayne, 2011): theoretical complexity ignores memory hierarchy effects that dominate runtime on modern hardware.

9. Conclusion

This study conducted a comparative theoretical and empirical analysis of four canonical sorting algorithms. The principal findings are:

- Quick Sort is the fastest algorithm on random input in practice, outperforming all alternatives by 1.6 to 2.1× due to cache-friendly access patterns and low constant factors.
- Merge Sort provides the most consistent cross-distribution performance and is recommended for applications requiring guaranteed O(n log n) performance regardless of input characteristics.
- Heap Sort's O(1) space advantage comes at a significant practical performance cost due to poor cache locality, making it the slowest of the three O(n log n) algorithms in all empirical tests.
- Insertion Sort is only competitive for very small inputs (n < ~30) or nearly-sorted data, where its $\Theta(n)$ best-case performance is superior to all alternatives.
- Empirical results consistently confirm theoretical complexity predictions, but reveal that constant factors specifically, primarily driven by CPU cache behaviour, produce substantial practical performance differences within the same asymptotic class.

These findings validate the approach taken by industrial sorting implementations (Timsort, Introsort) which hybridise algorithms to exploit the best properties of each: Quick Sort's average-case speed, Heap Sort's worst-case guarantee, and Insertion Sort's small-n efficiency. Future work could extend this study to include dual-pivot Quick Sort, Timsort, and radix sort, and could investigate cache miss rates using hardware performance counters for a more mechanistic explanation of the observed constant factor differences.

10. References

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